

Here is a comprehensive summary of our bench project and conversation context, broken down by major highlights and milestones:

Project Profile: Vintage L&R Watch Cleaning Unit Restoration

You are executing a complete mechanical, cosmetic, and electrical restoration of a vintage L&R Manufacturing Co. watch cleaning machine motor and basket assembly. The goal is to bring the machine back to factory-fresh, safely operating standards while modernizing critical components.

Housing Finish & Cosmetics

- **The Plan:** The cast-aluminum exterior motor housing halves are being stripped, prepped, and masked for refinishing.
- **The Coating:** You are applying a classic black industrial look using **VHT SP201 Wrinkle Finish spray paint** over a **Rust-Oleum 249322 Self-Etching Primer** base.
- **Curing Strategy:** Lacking a dedicated curing oven, the plan is to use a heat gun to initiate the wrinkle texture, followed by solar baking in the Southern California sun to fully harden the industrial coating.

Small Component & Basket Cleaning

- **The Mesh Basket Assembly:** The heavy-duty aluminum mesh cleaning basket, center column, and top impeller bracket were heavily fouled with decades of baked-on, dark chemical sludge and oxidation.
- **The Process:** You ran the assembly through an ultrasonic tank paired with a **Citranox** solution.
- **The Result:** The process completely stripped the scale, leaving the intricate mesh open, bright, and functionally pristine.

Gasket Conditioning & Preservation

- **Identification:** The main motor cap seal was verified to be a single, solid piece of molded rubber featuring original vintage patent text (U.S. PATENT 1872812).
- **De-Staining:** An initial soap scrub left behind an orange-brown chemical film caused by absorbed vintage cleaning solutions. A follow-up scrub with a warm **Citranox** solution successfully dissolved the surface film and made the patent text crisp and legible again.
- **Lubricant Warning:** We ruled out the use of WD-40 for cleaning, as its petroleum distillates swell, soften, and degrade vintage rubber elastomers.
- **Rejuvenation:** The gasket underwent a 16+ \text{ hour} immersion soak in **SEKODAY Pure Silicone Oil**. After a microfiber buffing technique, the rubber completely re-saturated, restoring its original structural pliability and locking in a rich, uniform satin-black finish.
- **Hardware Upgrade:** The original fastening system is being upgraded to a clean, modern screw-and-nut system to ensure even compression upon reassembly.



Sleeve Bearing & Oil Wick Overhaul

- **The Mechanism:** Teardown revealed that the motor utilizes **self-aligning spherical sleeve bearings** ("ball-and-socket" style) rather than standard fixed window blocks.
- **The Original Configuration:** The factory oiling reservoir used a continuous length of natural yarn threaded directly through a central lubrication hole to touch the shaft, with the remaining length wrapped tightly around the globe of the bearing. Flat wool felt washers were positioned at the edges of the housing pocket to block dust and contain oil migration.
- **Sourcing New Materials:** * *Reservoir Packing:* You selected **Estako Wool 98 (100% Superwash Merino Worsted Yarn, Off-White)** to replace the gunky factory yarn. The un-dyed superwash treatment ensures it won't leach dyes or break down into a hard mat when saturated with your **Zoom Spout** precision oil.
 - *Wick/Washer Stock:* You sourced **Singer Potted Motor F-1 Hard Wool Felt Grease Wicks** (1\text{ inch} long, trim-to-fit). These can be easily sliced down to cut fresh, custom-fit dust/retaining washers for the spherical bearing pockets.



Upcoming Phases

- **Rewiring & Electrical Modernization:** The original brittle wiring is slated for a complete replacement using continuous, highly flexible, silicone-coated wires.
- **Safety Grounding:** Because the original vintage design completely lacked grounding, you will be modifying the internal layout to integrate a true chassis ground to ensure safe operation by modern electrical standards.